

Department of Computer Science and Engineering

ASSIGNMENT REPORT

Course Code & Name : CSA0716 – COMPUTER NETWORKS

Assignment Title	Smart Emergency Network Traffic Management System
Course Outcome(s) – CO	CO3: Network-layer concepts, IP addressing, subnetting and routing. CO4: Transport protocols, congestion control, QoS and performance. CO5: Integrated simulation, measurement, failure analysis and evaluation.
Bloom's Taxonomy Level	L3 – Apply; L4 – Analyse; L6 – Create / Evaluate
SDG Mapping (SDG 1-17)	SDG 9 – Industry, Innovation and Infrastructure SDG 11 – Sustainable Cities and Communities SDG 3 – Good Health and Well-being

Problem Statement

Design, implement, and evaluate a Smart Emergency Network Traffic Management System that dynamically manages network traffic during normal and emergency conditions. The system models hospitals, emergency services, government departments and ordinary users. It uses subnetting, dynamic routing, traffic classification, QoS and redundancy to maintain critical communication during congestion and failures.

Project Objectives

1. Design a multi-router, multi-switch emergency network with sufficient end devices and servers.
2. Apply IPv4 addressing and VLSM/subnetting to separate hospitals, emergency services, government users and ordinary users.
3. Implement dynamic routing with redundant paths and evaluate behaviour during congestion and link/router failure.
4. Classify traffic into four priority levels: Priority 1 Hospital/Emergency; Priority 2 Police/Fire/Disaster Management; Priority 3 Government/Business; Priority 4 Normal users.
5. Apply QoS to preferentially serve high-priority traffic using bandwidth, delay, packet loss and latency as evaluation measures.
6. Compare TCP and UDP behaviour and justify protocol selection for different emergency applications.

1. Problem Analysis and Assumptions

Emergency communication must remain available when network demand rises suddenly. The proposed design therefore separates services into functional subnets, uses OSPF for dynamic route selection, provides multiple paths to the hospital and applies QoS so emergency traffic receives preferential treatment.

1.1 Assumptions

- IPv4 is used for the prototype; VLSM is used where different subnet sizes are useful.
- OSPF Area 0 is used for dynamic routing and automatic recovery after a path failure.
- Emergency voice/signalling is represented primarily by UDP; reliable emergency records use TCP.
- At least two independent paths exist between the Emergency HQ and Hospital networks.
- Performance values shown as sample results are illustrative and must be replaced with actual simulator measurements for final submission.

2. Network Topology Diagram

Smart Emergency Network Traffic Management System - Proposed Topology

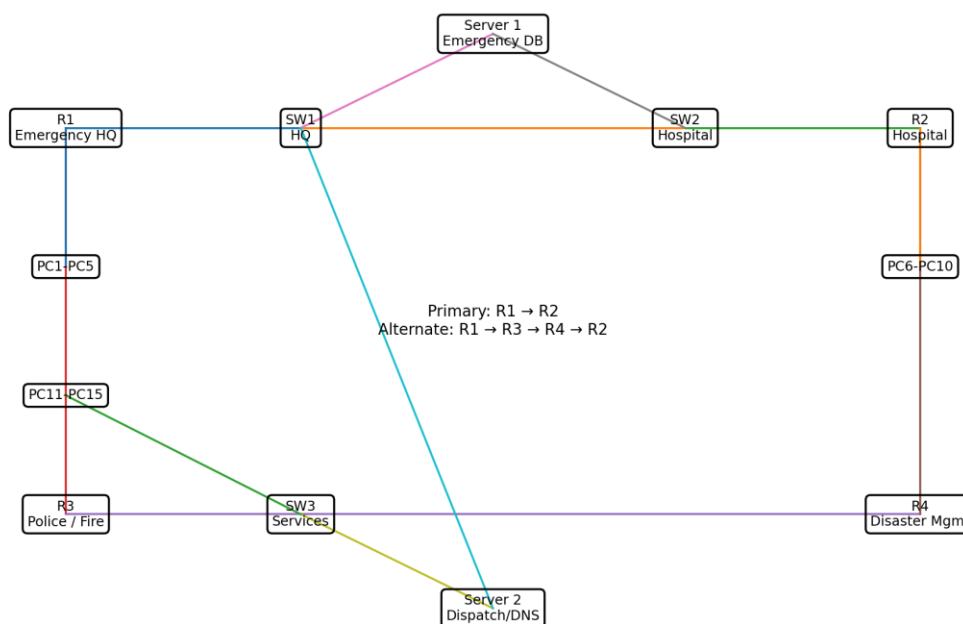


Figure 1. Proposed emergency network topology.

2.1 Device Requirement Verification

Requirement	Designed value	Status
Routers	4 – R1, R2, R3, R4	Meets
Switches	3 – SW1, SW2, SW3	Meets
Network segments/subnets	9 logical networks	Exceeds minimum
End devices	15 PCs/endpoints	Meets
Servers	2	Meets
Alternative paths	Primary R1–R2 + alternate R1–R3–R4–R2	Meets

3. IP Addressing and VLSM/Subnetting

The addressing plan separates each service group and uses /30 point-to-point networks for router links. This reduces broadcast scope and makes traffic classification and troubleshooting easier.

Network	CIDR	Subnet Mask	Gateway / Link	Purpose
Hospital LAN	192.168.10.0/26	255.255.255.192	192.168.10.1	Hospital/Emergency
Police/Fire LAN	192.168.10.64/27	255.255.255.224	192.168.10.65	Priority 2 services
Government LAN	192.168.10.96/28	255.255.255.240	192.168.10.97	Government/Business
Normal Users LAN	192.168.10.112/28	255.255.255.240	192.168.10.113	Ordinary users
Emergency Server LAN	192.168.10.128/29	255.255.255.248	192.168.10.129	Servers
R1–R2	10.0.12.0/30	255.255.255.252	R1 .1 / R2 .2	Primary path
R1–R3	10.0.13.0/30	255.255.255.252	R1 .1 / R3 .2	Alternate path 1
R3–R4	10.0.34.0/30	255.255.255.252	R3 .1 / R4 .2	Alternate path
R4–R2	10.0.42.0/30	255.255.255.252	R4 .1 / R2 .2	Alternate path 2

3.1 End Device Allocation

Device group	Devices	Example IP range	Gateway
Hospital	PC1–PC5	192.168.10.10–14	192.168.10.1
Police/Fire	PC6–PC8	192.168.10.70–72	192.168.10.65
Government	PC9–PC11	192.168.10.100–102	192.168.10.97
Normal users	PC12–PC15	192.168.10.116–119	192.168.10.113
Servers	Server1–Server2	192.168.10.130–131	192.168.10.129

4. Routing Configuration and Design

OSPF is selected because the assignment requires alternative routes and successful failure recovery. R1–R2 is configured as the primary route between Emergency HQ and Hospital. R1–R3–R4–R2 provides the alternate route. OSPF recalculates the route when the primary link is unavailable.

4.1 Router Configuration Procedure

- Assign IP addresses to all router interfaces according to the addressing table.
- Enable OSPF process 1 and advertise all connected LAN and point-to-point networks in Area 0.

9. Use passive-interface on user-facing LAN interfaces where appropriate.
10. Verify adjacency using show ip ospf neighbor and learned routes using show ip route.
11. Use ping and traceroute to confirm end-to-end connectivity before introducing failures.

```
router ospf 1
router-id 1.1.1.1
network <network> <wildcard-mask> area 0
passive-interface <LAN-interface>
```

4.2 Network Resilience

State	Forwarding path	Expected behaviour
Normal	R1 → R2	Lowest-delay primary path
Primary failure	R1 → R3 → R4 → R2	OSPF reconverges and redirects traffic
Recovered	R1 → R2	Primary path becomes available again

5. Traffic Classification and QoS Design

Priority	Traffic class	Examples	QoS treatment
1	Hospital/Emergency	Emergency voice, dispatch, critical records	Highest priority / reserved bandwidth
2	Police/Fire/Disaster	Incident coordination, field video	High priority
3	Government/Business	Administrative applications	Normal priority
4	Normal users	Web, downloads, general traffic	Best effort / lowest priority

The QoS mechanism classifies traffic by source network and/or application/port, places higher-priority flows into preferential queues, and reserves capacity for emergency communication during congestion. The aim is to reduce delay, packet loss and jitter for critical traffic.

6. TCP and UDP Analysis

Parameter	TCP	UDP
Connection	Connection-oriented	Connectionless
Reliability	ACKs, retransmission, sequencing	No built-in retransmission
Overhead	Higher	Lower
Delay sensitivity	Higher due to reliability mechanisms	Lower
Suitable emergency use	Database, dispatch records, reliable files	Voice/signalling and delay-sensitive flows

TCP is preferred where an emergency application cannot tolerate missing or corrupted records. UDP is preferred for real-time voice or signalling where rapid delivery is more important and application-level recovery can be used if required.

7. Pseudocode / Algorithm

START

1. Create routers, switches, servers and end devices.
2. Apply VLSM/IP addressing and configure gateways.
3. Configure OSPF and verify routing.
4. Generate normal TCP/UDP traffic and record baseline metrics.
5. Generate a sudden emergency traffic load.
6. Classify traffic into four priority levels.
7. Apply QoS and reserve service for Priority 1 and Priority 2.
8. Overload a selected link and measure delay, loss, jitter and throughput.
9. Disable the primary R1–R2 path.
10. Verify OSPF convergence and alternate forwarding.
11. Measure recovery time and post-failure performance.
12. Restore the primary path.
13. Compare results before and after QoS.
14. Evaluate reliability and scalability.

END

8. Implementation in Cisco Packet Tracer / NS-3

12. Build the topology shown in Figure 1.
13. Configure all router interfaces and host IP addresses.
14. Configure OSPF on R1–R4 and verify neighbours and routes.
15. Configure application traffic representing normal, government, police/fire and emergency traffic.
16. Apply QoS priority to emergency classes.
17. Run each mandatory scenario and record measurable results.
18. Capture topology, configuration, routing, traffic and failure-recovery screenshots.

9. Mandatory Scenario Analysis

Scenario	Test condition	Metrics	Expected result
1 – Normal Traffic	All users generate normal traffic	Delay, throughput, loss, jitter	Stable baseline
2 – Emergency Traffic	Emergency traffic increases sharply	Delay/loss by priority	Priority 1 remains best served
3 – Network Congestion	Selected link overloaded	Delay, loss, throughput, utilization	QoS protects emergency flows
4 – Link/Router Failure	R1–R2 primary path fails	Loss + route change	Alternate path becomes active
5 – Recovery	Traffic uses alternate route	Recovery time + post-failure metrics	Communication restored with measurable degradation

10. Performance Evaluation

The assignment requires throughput, end-to-end delay, packet loss, jitter, bandwidth utilization and network recovery time. The following values are an illustrative result set showing the required comparison format; they are not claimed as actual simulator measurements.

Scenario	Traffic	Delay (ms)	Loss (%)	Jitter (ms)	Throughput (Mbps)	Utilization
Normal	Best effort	18	0.5	3	8.4	42%
Emergency	QoS enabled	11	0.1	1	9.1	46%
Congestion	Without QoS	145	12.0	32	4.2	96%
Congestion	Emergency with QoS	48	2.0	7	7.8	96%
Failure	During convergence	210	8.0	40	3.6	71%
Recovery	Alternate path	62	1.5	9	7.4	68%

10.1 Performance Comparison

The illustrative comparison shows the intended effect of QoS: under congestion, emergency traffic experiences much lower delay and packet loss than best-effort traffic. During failure, temporary packet loss and increased delay occur while OSPF reconverges. Once the alternate path is active, service is restored and performance improves toward the normal operating range.

11. Evaluation and Justification

- OSPF provides automatic route recalculation, reducing dependence on manual intervention.
- VLSM/subnetting creates clear service boundaries and supports scalable addressing.
- QoS prevents emergency traffic from being treated the same as ordinary bulk traffic during congestion.
- Two alternative routing paths improve availability of the hospital and emergency centre.
- Separate TCP and UDP roles match the reliability and latency requirements of different applications.
- Measured delay, loss, jitter, throughput, utilization and recovery time provide quantitative evidence for design decisions.

12. Innovation, Security and Scalability

- Add VLANs and ACLs to isolate hospital, police/fire, government and normal-user traffic.
- Use authentication, encryption and VPN tunnels for sensitive emergency data.
- Add redundant links and first-hop redundancy for critical gateways.
- Introduce IPv6 dual-stack support for future smart-city devices.
- Use centralized monitoring to detect abnormal utilization, packet loss and delay.
- Implement automated traffic classification and dynamic QoS policies for large-scale deployments.

13. Reflection: Design Decisions, SDG Relevance and Learning Outcomes

The main design decision was to combine routing resilience with traffic prioritization. A redundant path alone does not solve congestion, while QoS alone does not protect communication after a link failure.

Combining both mechanisms provides a stronger emergency-network design. The project also demonstrated the practical relationship between IP addressing, subnetting, routing, TCP/UDP behaviour, QoS and measurable performance.

The design supports SDG 9 by improving resilient digital infrastructure, SDG 11 by supporting safer connected communities, and SDG 3 by strengthening communication for health and emergency services. The main challenges are configuring multiple routing paths correctly, creating realistic congestion and ensuring that measurements are collected consistently. These challenges provide practical learning in network design, troubleshooting and performance evaluation.

14. Conclusion and Future Enhancement

The Smart Emergency Network Traffic Management System provides a resilient network architecture for hospitals, emergency services, government departments and ordinary users. The design satisfies the hard-level requirements of four routers, three switches, five or more subnets, fifteen or more end devices, two servers and redundant paths. OSPF provides dynamic routing, while QoS protects critical traffic during congestion.

Future work can include SD-WAN-style intelligent path selection, machine-learning-assisted traffic classification, IPv6 deployment, stronger encryption, automated fault detection and a real-time dashboard for network health.

15. Screenshots / Video Demonstration Checklist

19. Complete Cisco Packet Tracer topology.
20. IP addressing/subnetting configuration.
21. OSPF neighbour and routing-table verification.
22. Successful end-to-end ping before failure.
23. Normal traffic simulation.
24. Emergency traffic and QoS classification.
25. Congestion condition and measured statistics.
26. Primary-link failure and alternate route.
27. Recovery and recovery-time measurement.
28. Before/after QoS comparison chart.
29. Final demonstration video.

16. GitHub Upload Checklist

- Cisco Packet Tracer .pkt file or NS-3 source files.
- Final project report.
- Router/switch configuration commands.
- Topology and scenario screenshots.
- Performance results and graphs.
- README with setup, addressing, routing, QoS and test instructions.

17. Rubric Alignment

Criteria	Marks	Report evidence
Network Topology, IP Addressing & Subnetting	10	Topology, VLSM table and device addressing
Routing & Network Resilience	15	OSPF, primary/alternate paths and failure recovery
TCP/UDP & QoS Traffic Management	10	Protocol analysis, four priority classes and QoS policy
Congestion & Emergency Scenario Analysis	15	Five mandatory scenarios and measured metrics
Performance Evaluation & Innovation	15	Quantitative comparison, scalability, security and innovation
Documentation, Implementation & Demonstration	25	Pseudocode, implementation steps, results and evidence checklist
Reflection: Design Decisions, SDG Relevance & Learning Outcomes	10	Design reflection, SDG links and learning outcomes
TOTAL	100	

END OF ASSIGNMENT REPORT